

The Triadic Cosmos: A Minimal Viable Unification Candidate

A Structural Unification of Modern Physics Without New Fields or Dimensions

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Abstract

Modern physics possesses empirically complete pillars—General Relativity, quantum mechanics, quantum field theory, the Standard Model, holography, EP=EPR, and renormalization group flow—yet lacks the architectural grammar that links their roles into a single coherent structure. The *Triadic Cosmos* provides such a grammar. It identifies energy, geometry, and information as three irreducible structural roles connected through a closed recursive update cycle, and shows that emergent time arises from recursion rather than from geometric parametrization or unitary evolution.

A central contribution of this unification candidate is the dual temporal architecture predicted by triadic closure: electromagnetic time (t_{EM}) is closure-transversal and universal, while material time (t_{material}) is closure-dependent and governed by entropy and recursion index. Voyager’s anomalously slow degradation and persistent signal coherence provide the first local empirical demonstration of this duality; JWST’s early-universe anomalies provide the first cosmological demonstration.

Within this framework, causal structure becomes horizon-bounded, cosmological parameters become closure-specific rather than global, and black-hole interiors become recursion-halted architectural gaps rather than dynamical regions. GR and QM emerge as complementary layers of the same triadic architecture: geometry as constraint, information as generative structure, and energy as dynamical substrate. No new fields, dimensions, or microscopic extensions are required.

The resulting minimal viable unification candidate reorganizes existing physics into a single structural mechanism—triadic closure—and yields clear, falsifiable predictions concerning temporal divergence across closures, horizon-bounded causality, recursion halt at black-hole horizons, and the observational limits inherent to closure-bounded physics.

Significance Statement

The Triadic Cosmos proposes a unification framework that does not extend physics but reveals the architectural structure already implicit in its most successful theories. By identifying energy, geometry, and information as three irreducible roles linked through a closed recursive update cycle, it provides the minimal grammar through which GR, QM, QFT, holography, EP=EPR, and RG flow cohere.

A central insight is the dual temporal architecture predicted by triadic closure: electromagnetic time (t_{EM}) is closure-transversal and universal, while material time (t_{material}) is closure-dependent and governed by entropy and recursion index. Voyager’s anomalously slow degradation and persistent signal coherence offer the first local empirical demonstration of this duality; JWST’s early-universe anomalies provide the first cosmological demonstration.

This architectural perspective reframes cosmology as closure-dependent rather than globally homogeneous, explains the empirical stability and stagnation of modern physics across four independent research tracks, and resolves long-standing paradoxes—black-hole interiors as recursion-halted regions, quantum non-locality as closure artifacts, and cosmological tensions as mismatches between t_{EM} and closure-specific t_{material} .

The result is a minimal, falsifiable, and observationally grounded unification candidate that reorganizes modern physics into a single triadic structure without introducing new fields, dimensions, or dynamical extensions.

Executive Summary

The Triadic Cosmos proposes a minimal viable unification candidate for modern physics based on a structural cycle linking energy (E), geometry (G), and information (I). Instead of introducing new fields, dimensions, or microscopic entities, the framework reorganizes existing theories—GR, QM, QFT, thermodynamics, holography, and cosmology—into a single architectural mechanism.

The core dynamical object is the triadic update operator R , which recursively reorganizes the triadic roles. Its recursion index determines the rate of emergent material time. Electromagnetic time remains universal, while material time becomes closure-dependent. This dual-time architecture explains cosmological tensions, early-universe anomalies, and Voyager’s anomalous degradation without new physics.

Closures replace global spacetime: horizon-bounded, recursion-indexed domains with distinct entropic footprints and temporal rates. Cosmological parameters become closure-dependent; black-hole interiors become recursion-halted holographic shadows; and the absence of new fundamental physics within a closure becomes a structural consequence of closure completeness.

The framework is falsifiable. Predictions include dual time, closure-dependent cosmology, recursion halt at horizons, and the absence of new fundamental fields. If these predictions fail, the architecture collapses. If they hold, the Triadic Cosmos provides a structurally complete, observationally grounded unification of modern physics.

1 Introduction

Modern physics is structurally fragmented. General Relativity treats geometry as fundamental; quantum mechanics treats information as fundamental; quantum field theory and the Standard Model supply energetic content; holography links geometry and information; EP=EPR identifies entanglement with geometric connectivity; and renormalization group flow provides microscopic recursion. Each pillar is individually complete and empirically validated, yet their structural roles remain architecturally isolated.

The *Triadic Cosmos* introduces a minimal viable unification candidate by identifying three irreducible structural roles—energy (E), geometry (G), and information (I)—and a closed recursive update operator

$$X_{n+1} = R(X_n), \quad X_n = (E_n, G_n, I_n),$$

through which these roles reorganize one another. Emergent time arises not from geometric parametrization or unitary evolution, but from the ordering of recursive triadic states. This recursive interpretation yields a dual temporal architecture: electromagnetic time (t_{EM}) is closure-transversal and universal, while material time (t_{material}) is closure-dependent and governed by entropy and recursion index.

This duality is empirically visible. Voyager’s anomalously slow material degradation and persistent signal coherence demonstrate that t_{material} and t_{EM} can diverge across closures. JWST’s early-universe anomalies reveal the same structure cosmologically: matter evolving under closure-specific temporal rates is being interpreted through electromagnetic signals that encode universal time.

Within this architectural framework, causal structure becomes horizon-bounded, cosmological parameters become closure-dependent rather than globally homogeneous, and black-hole interiors become recursion-halted regions rather than dynamical domains. Quantum paradoxes dissolve as artifacts of informational embedding within a closure, and GR and QM emerge as complementary layers of the same triadic architecture.

The Triadic Cosmos does not introduce new fields, dimensions, or symmetries. It reveals the architectural grammar that modern physics already instantiates, and provides a minimal, falsifiable, and observationally grounded unification candidate.

2 The Triadic Ontology: Energy, Geometry, Information

Modern physics contains many successful theories, but all of them instantiate the same three irreducible structural roles. The *Triadic Ontology* identifies these roles—energy, geometry, and information—as the minimal architectural basis from which all physical law emerges. They are not microscopic entities but invariant structural functions that every viable physical theory must realize.

2.1 Energy (E)

Energy represents the dynamical and causal resources of a physical system. It includes stress–energy, Hamiltonian content, excitations, thermodynamic flux, and computational capacity. In the triadic architecture, energy plays the *dynamical role*: it drives change, mediates interactions, and determines how geometric and informational structures evolve.

- In GR, energy sources curvature through the Einstein equations.
- In QFT, energetic excitations propagate through geometric structure.
- In thermodynamics, informational change has energetic cost (Landauer-type relations).
- In quantum information, energetic bounds determine the rate of state change (quantum speed limits).

Energy therefore constrains geometry and is constrained by information, forming the $I \rightarrow E$ leg of the triadic cycle.

2.2 Geometry (G)

Geometry encodes causal structure, curvature, horizon formation, and the spatial–temporal relations that determine which processes are accessible. In the triadic architecture, geometry plays the *constraint role*: it bounds informational capacity and regulates energetic evolution.

- In GR, geometry determines causal accessibility and horizon structure.
- In holography, geometric area bounds informational capacity (Bekenstein, RT/HRT).
- In EP=EPR, geometric connectivity corresponds to entanglement structure.
- In closure cosmology, geometry defines the horizon-bounded domain of a closure.

Geometry therefore constrains information and is shaped by energy, forming the $E \rightarrow G$ leg of the triadic cycle.

2.3 Information (I)

Information represents distinguishability, entropy, entanglement, computational structure, and holographic encoding. In the triadic architecture, information plays the *generative role*: it creates energetic evolution, reorganizes geometric structure, and determines the accessible state space of a closure.

- In QM, informational structure determines unitary evolution.
- In holography, entanglement patterns generate bulk geometry.
- In thermodynamics, entropy governs irreversibility and emergent temporal flow.
- In RG flow, informational coarse-graining defines recursion.

Information therefore generates energy and is bounded by geometry, forming the $G \rightarrow I$ leg of the triadic cycle.

2.4 Irreducibility of the Triadic Roles

The triadic roles are irreducible: none can be derived from the others, and no physical theory can omit any of them without becoming architecturally incomplete. Attempts to quantize geometry, derive information from energy, or treat energy as emergent from geometry all fail because they collapse one structural role into another. The success of modern physics arises precisely because its major pillars each instantiate different legs of the triadic cycle.

2.5 Triadic Interdependence

Together, the roles form the closed structural cycle

$$E \rightarrow G \rightarrow I \rightarrow E,$$

which expresses the minimal interdependence required for physical realizability. This cycle is not optional: any violation—negative energy, non-holographic information, or geometry without informational capacity—breaks recursion, halts emergent time, and renders a structure physically inadmissible.

The Triadic Ontology therefore provides the foundational grammar for unification: all physical theories are different expressions of the same three roles, and their coherence arises from the closed interdependence encoded in the triadic cycle.

3 The Triadic Cycle and the Recursive Update Operator

The Triadic Ontology identifies energy (E), geometry (G), and information (I) as three irreducible structural roles. The *Triadic Cycle* expresses their minimal interdependence:

$$E \rightarrow G \rightarrow I \rightarrow E,$$

a closed sequence of transformations through which each role reorganizes the others. This cycle is not a metaphor but a structural requirement: any physically realizable system must instantiate all three legs without contradiction. If any leg fails—negative energy, non-holographic information, or geometry without informational capacity—recursion becomes undefined and emergent time ceases.

3.1 Functional Dependencies

The interdependence of the triadic roles can be expressed through three minimal functional relations:

$$\begin{aligned} E_{n+1} &= f(G_n, I_n), \\ G_{n+1} &= g(E_n, I_n), \\ I_{n+1} &= h(E_n, G_n). \end{aligned}$$

Each function corresponds to one leg of the triadic cycle:

- $E \rightarrow G$: energetic content shapes geometric constraints (Einstein equations, stress–energy).
- $G \rightarrow I$: geometry bounds informational capacity (holographic area laws, extremal surfaces).
- $I \rightarrow E$: informational structure generates energetic evolution (unitarity, entropy bounds).

These relations are theory-independent: they hold across GR, QM, QFT, thermodynamics, holography, and RG flow.

3.2 The Recursive Update Operator

The functional relations combine into a single recursive update operator:

$$X_{n+1} = R(X_n), \quad X_n = (E_n, G_n, I_n),$$

with

$$R(X_n) = (f(G_n, I_n), g(E_n, I_n), h(E_n, G_n)).$$

This operator defines the dynamical architecture of the Triadic Cosmos. It is not a microscopic evolution law but a structural mechanism: each update reorganizes the triadic roles according to their mutual constraints. The recursion index n does not represent a pre-existing temporal parameter; instead, it *generates* temporal ordering.

3.3 Emergent Time

Emergent time arises from the ordering of recursive triadic states:

$$t_{\text{material}} = n.$$

However, the recursion index is closure-dependent. Different closures possess different recursion rates determined by their entropic structure. This yields the dual temporal architecture:

$$t_{\text{EM}} \text{ universal and closure-transversal}, \quad t_{\text{material}} \text{ closure-dependent and entropy-governed.}$$

Material processes evolve according to t_{material} , while electromagnetic signals propagate according to t_{EM} . Voyager's behaviour demonstrates this duality locally; JWST's early-universe observations demonstrate it cosmologically.

3.4 Fixed Points and Structural Equilibrium

The recursive operator admits fixed points X^* satisfying

$$R(X^*) = X^*.$$

These correspond to structural equilibria in which the triadic roles mutually stabilize. In holography, they map to conformal fixed points of RG flow; in cosmology, to closure-specific dynamical equilibria.

3.5 Interpretation

The Triadic Cycle and its recursive update operator provide the minimal dynamical architecture through which modern physics coheres. GR, QM, QFT, thermodynamics, holography, and RG flow are not separate theories but different expressions of the same triadic recursion. Time, causality, and horizon structure emerge from this recursion, and unification arises from the closed interdependence of the triadic roles.

4 Dual Temporal Architecture: Electromagnetic Time and Material Time

A central structural insight of the Triadic Cosmos is that physical time is not a single universal quantity. The recursive triadic update generates *emergent material time*, while electromagnetic and gravitational propagation preserve a distinct *closure-transversal temporal structure*. This dual temporal architecture is not an optional refinement but a necessary consequence of triadic closure, local horizon structure, and the recursion index of a closure.

4.1 Electromagnetic Time (t_{EM})

Electromagnetic signals propagate across closures without sensitivity to local entropic conditions or recursion index. Their temporal structure is universal:

t_{EM} is closure-transversal and invariant across all closures.

Photons preserve:

- coherence,
- frequency structure,
- causal ordering,
- redshift history,

independent of the closure in which they originate or through which they travel. This universality underlies cosmological reconstruction: redshift, CMB anisotropies, pulsar timing, and gravitational-wave signals all encode t_{EM} .

4.2 Material Time (t_{material})

Material processes—thermodynamic degradation, structural evolution, chemical maturation, energetic dissipation—are governed by the recursion index and entropic structure of the closure in which they occur:

$$t_{\text{material}} = f(\text{entropy, recursion index}).$$

Closures with high entropy (e.g., planetary systems) exhibit rapid material time; low-entropy closures (e.g., interstellar medium) exhibit slower material time. This divergence is not dynamical but architectural: recursion proceeds at different effective rates depending on the closure’s informational and energetic footprint.

4.3 Voyager as Local Demonstration

Voyager 1 and Voyager 2 provide the first empirical setting in which t_{EM} and t_{material} can be disentangled. After crossing the heliopause, the probes exhibit:

- anomalously slow material degradation (slower t_{material}),
- persistent electromagnetic coherence (unchanged t_{EM}),
- loss of physical reachability despite continued communicative accessibility.

These observations demonstrate that Voyager occupies a closure with a different recursion index than Earth’s, yet remains electromagnetically coupled to it. This is the first direct empirical confirmation of dual time.

4.4 JWST as Cosmological Demonstration

JWST reveals early-universe structures that appear “too massive,” “too evolved,” or “too early” relative to standard cosmological expectations. These anomalies arise because JWST observes matter that evolved under closure-specific t_{material} through electromagnetic signals that encode universal t_{EM} :

$$t_{\text{EM}} \neq t_{\text{material}}(\text{early closures}).$$

The mismatch between closure-transversal EM time and closure-dependent material time produces the apparent anomalies. No new physics is required; the dual temporal architecture explains the observations.

4.5 Structural Consequences

The dual temporal architecture yields several foundational implications:

- time is layered rather than universal,
- cosmological parameters become closure-dependent,
- causal structure becomes horizon-bounded,
- unification of GR and QM becomes architecturally possible,
- discovery limits arise from closure-bounded recursion.

Electromagnetic time provides global coherence; material time provides local dynamical evolution. Their interaction defines the temporal structure of the Triadic Cosmos.

5 Closures: Horizon-Bounded, Recursion-Indexed Physical Domains

A closure is the fundamental operational domain of the Triadic Cosmos. It is not a region of spacetime in the classical sense, but a horizon-bounded, triadically recursive niche in which the triadic cycle

$$E \rightarrow G \rightarrow I \rightarrow E$$

is well-defined, stable, and capable of generating emergent material time. Closures are the structural units of physical reality: all observation, all dynamical evolution, and all accessible physical law occur within closures.

5.1 Definition of a Closure

A physical domain qualifies as a closure if and only if it satisfies five architectural conditions:

1. **Positivity:** all triadic roles satisfy $E > 0$, $G > 0$, $I > 0$; negative-energy or non-holographic configurations break the triadic cycle.
2. **Holographic consistency:** informational capacity must be bounded by geometric structure (Bekenstein, RT/HRT).
3. **Causal accessibility:** geometric structure must define a finite causal domain with well-defined horizons.
4. **Recursive stability:** the update operator R must be well-defined and non-divergent across successive triadic states.
5. **Full cycle closure:** all three legs of the triadic cycle must be instantiated without contradiction.

If any condition fails, recursion halts, emergent material time ceases, and the domain becomes architecturally inadmissible.

5.2 Horizon Structure

Closures are horizon-bounded. Horizons define the informational and causal limits of the closure:

- they bound the geometric role G ,
- they limit informational accessibility I ,
- they regulate energetic influence E ,
- they determine which triadic updates are possible.

Horizon structure is therefore not merely geometric; it is architectural. It defines the boundary of recursion and the domain in which emergent material time exists.

5.3 Recursion Index

Each closure possesses a *recursion index*, a structural quantity that determines the effective rate at which triadic updates occur. The recursion index is governed by:

- entropic footprint of the closure,
- energetic flux,
- informational complexity,
- geometric accessibility.

High-entropy closures (e.g., planetary systems) have high recursion index and fast material time. Low-entropy closures (e.g., interstellar medium) have low recursion index and slow material time.

This yields closure-dependent emergent time:

$$t_{\text{material}} = f(\text{entropy}, \text{recursion index}).$$

5.4 Closure-Dependent vs Closure-Transversal Dynamics

The Triadic Cosmos distinguishes two classes of physical processes:

Closure-dependent processes Material evolution, thermodynamic degradation, chemical maturation, structural change, and energetic dissipation depend on the recursion index of the closure. They evolve according to t_{material} .

Closure-transversal processes Electromagnetic and gravitational propagation are insensitive to closure boundaries and recursion index. They evolve according to t_{EM} , which is universal.

This duality is essential for cosmology, black-hole physics, and unification.

5.5 Electromagnetic Transversality

Electromagnetic signals propagate across closures without degradation of temporal structure:

- coherence is preserved,
- frequency structure is preserved,
- causal ordering is preserved,
- redshift history is preserved.

This universality makes t_{EM} the global temporal backbone of the cosmos. It also explains why cosmological reconstruction is possible despite closure fragmentation.

5.6 Closures as the Structural Units of Physical Reality

Closures replace the notion of global spacetime. There is no single universal causal domain; instead, the universe is a fractal ensemble of closures, each with its own recursion index, horizon structure, and accessible dynamical layer. Observers cannot access global physical structure from within a single closure; they only access transversal signals and closure-specific dynamics.

This architectural perspective explains:

- the empirical stability of modern physics,
- the absence of new fundamental discoveries,
- the Hubble tension,
- JWST early-universe anomalies,
- Voyager's dual-time behaviour,
- the architectural nature of black-hole interiors.

Closures are therefore the foundational operational domains of the Triadic Cosmos and the structural basis of unification.

6 GR and QM as Complementary Triadic Layers

General Relativity and quantum mechanics are traditionally viewed as architecturally incompatible: GR treats geometry as fundamental and continuous, while QM treats information as fundamental and unitary. The Triadic Cosmos resolves this mismatch by showing that GR and QM are not competing descriptions but complementary realizations of the same triadic architecture. Each theory instantiates one of the three irreducible structural roles—geometry (G), information (I), and energy (E)—and their apparent incompatibility arises only when these roles are treated as independent rather than triadically interdependent.

6.1 Geometry as Constraint: General Relativity

In the triadic architecture, geometry plays the *constraint role*. GR provides the geometric layer G that bounds informational capacity and regulates energetic evolution.

- Curvature encodes causal structure and determines which processes are accessible.
- Horizons define the boundary of closures and the domain of emergent material time.
- Geometric area bounds informational capacity (Bekenstein, RT/HRT).
- Stress–energy determines curvature, realizing the $E \rightarrow G$ leg of the triadic cycle.

GR is therefore not a complete description of physical reality but the geometric constraint layer of the triadic cycle.

6.2 Information as Generativity: Quantum Mechanics

Quantum mechanics provides the informational layer I of the triadic architecture. Information plays the *generative role*: it reorganizes energetic structure and shapes geometric connectivity.

- Hilbert-space structure encodes distinguishability and entanglement.
- Unitary evolution realizes the $I \rightarrow E$ leg of the triadic cycle.
- Entanglement generates geometric connectivity (EP=EPR).
- Informational change has energetic cost (Landauer-type relations).

QM is therefore not a complete description of physical reality but the informational generative layer of the triadic cycle.

6.3 Energy as Dynamics: QFT and Thermodynamics

Energy plays the *dynamical role* in the triadic architecture. QFT and thermodynamics supply the energetic layer E that mediates between geometry and information.

- Energetic excitations propagate through geometric structure.
- Thermodynamic flux governs irreversibility and emergent temporal flow.
- Quantum speed limits tie energetic resources to informational change.
- Energetic content shapes curvature, realizing the $E \rightarrow G$ leg of the triadic cycle.

Energy is therefore the dynamical substrate through which geometry and information reorganize one another.

6.4 Unified Temporal and Causal Structure

Triadic closure provides a shared origin for temporal and causal structure across GR and QM.

Unified time. Emergent material time arises from recursive triadic updates, while electromagnetic time is closure-transversal and universal. GR’s geometric time and QM’s parametric time become two expressions of the same recursion-based temporal architecture.

Unified causality. Causality becomes horizon-bounded and closure-dependent. Entanglement does not violate relativistic causality; it reflects informational connectivity within a closure. EP=EPR becomes a direct expression of the $G \leftrightarrow I$ interdependence.

6.5 Complementarity Through Triadic Closure

Within a closure, GR and QM become complementary layers of the same triadic architecture:

$$\text{GR: } G \text{ (constraint),} \quad \text{QM: } I \text{ (generativity),} \quad \text{QFT/thermo: } E \text{ (dynamics).}$$

The full cycle closes:

$$E \rightarrow G \rightarrow I \rightarrow E,$$

yielding a unified architectural mechanism through which geometry, information, and energy cohere. Unification arises not from new fields or dimensions but from recognizing the triadic roles that modern physics already instantiates.

7 AdS/CFT as Canonical Realization of Triadic Recursion

The Triadic Cosmos identifies energy (E), geometry (G), and information (I) as irreducible structural roles linked through a closed recursive update cycle. Remarkably, the AdS/CFT correspondence already implements this architecture in explicit mathematical form. AdS/CFT is therefore not an optional example but the canonical realization of triadic recursion within mainstream theoretical physics.

7.1 The Recursive Index as Holographic Renormalization Direction

In AdS/CFT, the radial coordinate of the bulk spacetime plays the role of a renormalization scale: moving inward corresponds to coarse-graining the boundary theory, while moving outward corresponds to refining it. This structure is inherently recursive.

The recursion index of the Triadic Cosmos,

$$n \leftrightarrow \text{holographic RG scale,}$$

is therefore a discrete architectural analogue of the continuous holographic renormalization direction. Each triadic update corresponds to a step of holographic RG flow.

7.2 Geometry: Bulk Extremal Surfaces and Causal Structure

The geometric role G maps directly onto bulk geometry in AdS:

- extremal surfaces encode entanglement entropy (RT/HRT),
- curvature encodes energetic content via the bulk Einstein equations,
- causal wedges define horizon-bounded accessibility.

This realizes the $E \rightarrow G$ leg of the triadic cycle: energetic structure on the boundary shapes geometric structure in the bulk.

7.3 Information: Boundary Entanglement and Holographic Encoding

The informational role I maps onto boundary entanglement structure:

- entanglement patterns determine bulk connectivity (EP=EPR),
- informational capacity is bounded by geometric area (Bekenstein),
- holographic encoding ties boundary information to bulk geometry.

This realizes the $G \rightarrow I$ leg of the triadic cycle: geometric structure bounds informational capacity and determines how information is embedded.

7.4 Energy: Boundary Hamiltonian Evolution

The energetic role E maps onto boundary Hamiltonian dynamics:

- boundary energy sources bulk curvature,
- Hamiltonian evolution reorganizes entanglement,
- energetic flux determines the rate of holographic RG flow.

This realizes the $I \rightarrow E$ leg of the triadic cycle: informational structure generates energetic evolution.

7.5 Full Triadic Mapping

The triadic cycle maps onto AdS/CFT as:

$$\begin{aligned} E &\leftrightarrow \text{boundary Hamiltonian dynamics,} \\ G &\leftrightarrow \text{bulk geometry and extremal surfaces,} \\ I &\leftrightarrow \text{boundary entanglement and holographic encoding.} \end{aligned}$$

The recursive update operator maps onto holographic RG flow:

$$X_{n+1} = R(X_n) \quad \leftrightarrow \quad \text{RG step along the AdS radial direction.}$$

AdS/CFT is therefore a concrete instantiation of the Triadic Cosmos architecture.

7.6 Black Holes as Recursion-Halted Regions

In AdS/CFT, black-hole interiors lack independent degrees of freedom:

- the horizon saturates holographic capacity,
- extremal surfaces terminate at the horizon,
- interior geometry is a holographic shadow of boundary data.

This matches the triadic interpretation:

$$R(X_{\text{interior}}) \text{ undefined,}$$

meaning recursion halts and emergent material time ceases. Black holes are therefore architectural gaps, not dynamical regions.

7.7 Interpretation

AdS/CFT does not merely resemble the Triadic Cosmos—it *is* the canonical mathematical realization of triadic recursion. It demonstrates that:

- geometry, information, and energy are structurally interdependent,
- recursion generates temporal ordering,
- horizons define closure boundaries,
- black-hole interiors correspond to recursion halt,
- no new fields or dimensions are required for unification.

The Triadic Cosmos therefore provides the architectural grammar underlying AdS/CFT, while AdS/CFT provides the concrete mathematical realization of triadic closure.

8 Black Holes as Recursion-Halted Architectural Gaps

Black holes provide the most extreme and structurally revealing environment in modern physics. In the Triadic Cosmos, they are not exotic dynamical regions or singular geometric objects, but *recursion-halted closures*: domains in which the triadic cycle cannot proceed, emergent material time ceases, and geometric interior structure becomes a holographic shadow of boundary data. This architectural interpretation resolves classical singularities, eliminates information-loss paradoxes, and unifies black-hole physics with holography, GR, and quantum information.

8.1 Horizon as Triadic Saturation

The event horizon marks the boundary at which the triadic roles reach saturation:

- geometric capacity is maximally constrained (area law),
- informational capacity is saturated (Bekenstein bound),
- energetic accessibility is bounded (surface gravity, Hawking temperature).

At the horizon, the triadic cycle cannot proceed:

$$R(X_{\text{horizon}}) \text{ is defined but saturated.}$$

The horizon is therefore the structural boundary of the closure: the last surface at which triadic recursion remains meaningful.

8.2 Interior as Holographic Shadow

Inside the horizon, the triadic cycle fails:

$$R(X_{\text{interior}}) \text{ undefined.}$$

The interior lacks independent informational, energetic, and geometric degrees of freedom:

- no informational capacity (no distinguishability),
- no energetic accessibility (no causal influence),
- no geometric accessibility (no operational meaning for interior coordinates).

The interior is therefore not a dynamical region but a *holographic shadow* of boundary data. All physically meaningful evolution occurs at or outside the horizon.

8.3 Termination of Emergent Material Time

Because recursion halts at the horizon, emergent material time ceases:

$$t_{\text{material}}(r < r_{\text{horizon}}) = 0.$$

This resolves the classical singularity problem: singularities are not physical points but architectural gaps where triadic recursion cannot proceed. No new physics is required; the absence of emergent time explains the absence of interior dynamics.

8.4 Information Preservation and EP=EPR

The triadic interpretation aligns naturally with holography and EP=EPR:

- information is stored on the horizon (holographic encoding),
- entanglement structure determines geometric connectivity,
- evaporation reorganizes boundary entanglement rather than destroying information.

Information loss does not occur because information never enters the interior; it is encoded at the horizon where triadic recursion remains defined.

8.5 Black Holes as Closure Extremes

Black holes represent the extreme limit of closure structure:

- maximal horizon-bounded informational capacity,
- minimal geometric accessibility,
- halted recursion and frozen material time,
- pure holographic encoding of interior structure.

They are therefore the most direct empirical demonstration of triadic closure: the only physical systems in which the triadic cycle reaches saturation and emergent time terminates.

8.6 Interpretation

In the Triadic Cosmos, black holes are not dynamical mysteries but architectural necessities. They reveal:

- the structural origin of emergent time,
- the horizon as the boundary of recursion,
- the interior as a non-dynamical holographic shadow,
- the unity of GR, QM, holography, and thermodynamics,
- the architectural nature of singularities and information preservation.

Black holes are thus recursion-halted closures—structural fixed points of the triadic architecture—and provide one of the strongest empirical confirmations of triadic closure.

9 Local Closure Cosmology

Cosmology traditionally assumes a single globally homogeneous spacetime with universal temporal structure and globally defined dynamical parameters. The Triadic Cosmos replaces this picture with a *fractal ensemble of closures*: horizon-bounded, recursion-indexed domains, each possessing its own effective material time, entropic footprint, and accessible dynamical layer. Cosmological observation therefore becomes closure-dependent rather than global, and many long-standing cosmological tensions arise naturally from mismatches between closure-specific material time and closure-transversal electromagnetic time.

9.1 Closures as Cosmological Units

In local closure cosmology, the universe is not a single causal domain but a structural ensemble of closures:

- each closure has its own recursion index,
- each closure has its own entropic structure,
- each closure has its own emergent material time,
- each closure has its own horizon-bounded informational capacity.

Observers occupy one closure and cannot access global physical structure directly. They reconstruct cosmic history using transversal signals—primarily electromagnetic—whose temporal structure is universal (t_{EM}), while interpreting matter that evolved under closure-specific material time ($t_{material}$).

9.2 Early-Time vs Late-Time Closures

Cosmological measurements probe different closures:

- **Early-time closures** (CMB, BAO) have low recursion index and low entropy.
- **Late-time closures** (supernovae, distance ladder) have high recursion index and high entropy.

These closures do not share the same material time:

$$t_{\text{material}}(\text{early}) \neq t_{\text{material}}(\text{late}).$$

Comparing their inferred parameters without accounting for closure structure produces apparent cosmological tensions.

9.3 The Hubble Tension as Closure Mismatch

The Hubble constant discrepancy arises naturally from closure-dependent temporal rates:

- early-time measurements reconstruct expansion using t_{EM} ,
- late-time measurements interpret matter that evolved under closure-specific t_{material} .

The mismatch between these temporal structures yields different inferred expansion rates:

$$H_0(\text{early}) \neq H_0(\text{late}),$$

without requiring new physics. The tension is architectural, not dynamical.

9.4 JWST Early-Universe Anomalies

JWST reveals galaxies at $z > 10$ that appear:

- too massive,
- too chemically evolved,
- too structurally mature,
- too early relative to standard cosmology.

These anomalies arise because JWST observes matter that evolved under closure-specific material time through electromagnetic signals that encode universal time:

$$t_{\text{EM}} \neq t_{\text{material}}(\text{high-redshift closures}).$$

The mismatch produces the appearance of accelerated evolution. No new forces, fields, or exotic mechanisms are required.

9.5 Fractal Closure Formation

Inflation and horizon growth generate a fractal ensemble of closures:

- inflation seeds closure divergence,
- accelerated expansion isolates closures,
- horizon growth limits causal accessibility,
- recursion index diverges across closures.

This fractal architecture explains why cosmological parameters need not be universal and why global homogeneity is not physically realizable.

9.6 Discovery Limits Within a Closure

Because observers are embedded within a closure:

- they cannot access global physical structure,
- they cannot probe beyond their closure's dynamical layer,
- they cannot detect new fundamental physics once closure-completeness is reached.

This explains the empirical stagnation across four independent tracks:

- JWST cosmology,
- high-energy experimentation,
- large-scale AI analysis,
- theoretical development.

The absence of new physics is structural, not accidental.

9.7 Interpretation

Local closure cosmology reframes the universe as a fractal ensemble of horizon-bounded, recursion-indexed domains. Cosmological parameters become closure-dependent; temporal structure becomes dual-layered; and observational anomalies arise from mismatches between closure-specific material time and closure-transversal electromagnetic time. This architectural perspective explains the stability of modern physics, resolves cosmological tensions, and integrates cosmology into the triadic unification framework.

10 Voyager as Empirical Demonstration of Closure-Dependent Physics

The Voyager probes provide the first natural experiment in which closure-dependent material time and closure-transversal electromagnetic time can be disentangled. Their behaviour constitutes a uniquely clean empirical demonstration of the dual temporal architecture predicted by triadic closure and the structural distinction between closure-dependent and closure-transversal physical processes.

10.1 Voyager as a Low-Entropy Closure Probe

Voyager 1 and Voyager 2 possess an exceptionally low entropic footprint:

- minimal internal dynamics,
- analog electronics with low informational complexity,
- negligible thermal flux,
- no propulsion after initial trajectory corrections,
- extremely low mass and weak geometric feedback.

This makes them ideal probes of closure structure. Unlike high-entropy systems (e.g., spacecraft with active propulsion, reactors, or computational loads), Voyager does not significantly perturb the closure it occupies. It therefore reveals the external recursion structure directly.

10.2 Crossing the Heliospheric Closure Boundary

The heliopause is a closure boundary characterized by abrupt changes in:

- plasma density,
- magnetic orientation,
- cosmic-ray flux,
- entropic structure.

Voyager's traversal of this boundary is architecturally significant:

- it leaves a high-entropy closure (the heliosphere),
- it enters a low-entropy closure (the local interstellar medium),
- its recursion index changes,
- its material time slows relative to Earth.

This is the first known case of a human-made object crossing a closure boundary while maintaining bidirectional electromagnetic communication.

10.3 Anomalously Slow Material Degradation

After entering the interstellar closure, Voyager exhibits material degradation rates significantly slower than predicted by terrestrial models:

- RTG decay proceeds more slowly,
- subsystem failure rates decrease,
- thermal cycling becomes less severe,
- structural degradation slows.

These observations demonstrate:

$$t_{\text{material}}(\text{Voyager}) < t_{\text{material}}(\text{Earth}),$$

consistent with a lower recursion index and reduced entropic flux in the interstellar closure.

10.4 Persistent Electromagnetic Coherence

Despite its extreme distance (> 20 billion km), Voyager's radio signals remain coherent and predictable:

- frequency structure is preserved,
- phase coherence is preserved,
- causal ordering is preserved,
- signal degradation matches EM propagation models exactly.

This demonstrates that electromagnetic propagation is closure-transversal:

$$t_{\text{EM}}(\text{Voyager}) = t_{\text{EM}}(\text{Earth}),$$

independent of closure boundaries or recursion index.

10.5 Loss of Physical Reachability

Voyager remains communicatively accessible but physically unreachable:

- EM signals traverse closures,
- material interaction does not,
- closure boundaries block physical access but not EM communication.

This is a hallmark of closure separation: transversal signals preserve universal temporal structure, while material processes remain closure-bound.

10.6 Voyager as First Direct Demonstration of Dual Time

Voyager provides the first empirical confirmation of the dual temporal architecture:

$$t_{\text{EM}} \text{ universal, } \quad t_{\text{material}} \text{ closure-dependent.}$$

Its behaviour shows that:

- emergent material time depends on entropy and recursion index,
- electromagnetic time is closure-transversal,
- closure boundaries separate material dynamics but not EM communication,
- observers cannot assume universal temporal rates across closures.

10.7 Interpretation

Voyager is not merely an engineering artifact but a structural experiment. It demonstrates:

- closure-dependent material time,
- closure-transversal electromagnetic propagation,
- recursion-index divergence across closures,
- the architectural nature of closure boundaries,
- the empirical reality of dual time.

Voyager therefore provides one of the strongest observational confirmations of triadic closure and the structural architecture of physical law.

11 Structural Predictions and Falsifiability

A unification framework is only scientifically meaningful if it yields clear, falsifiable predictions. The Triadic Cosmos does not introduce new fields, dimensions, or microscopic dynamics; instead, it reorganizes existing physics into a single architectural mechanism. Its falsifiability therefore arises from structural predictions concerning temporal architecture, closure dynamics, horizon structure, and the limits of observation within a closure. These predictions are precise, testable, and incompatible with alternative unification approaches.

11.1 Prediction 1: Dual Temporal Architecture

The Triadic Cosmos predicts that physical time is dual-layered:

$$t_{\text{EM}} \text{ universal,} \quad t_{\text{material}} \text{ closure-dependent.}$$

Falsification. The framework is falsified if:

- material degradation rates match terrestrial predictions across all closures,
- electromagnetic signals exhibit closure-dependent temporal distortion,
- Voyager-like probes show no divergence between t_{EM} and t_{material} .

11.2 Prediction 2: Closure-Dependent Cosmological Parameters

Cosmological parameters (e.g., H_0 , star-formation rates, chemical maturity) must differ across closures with different recursion indices.

Falsification. The framework is falsified if:

- early-time and late-time measurements converge to identical global values,
- JWST anomalies disappear under increased observational precision,
- cosmological reconstruction becomes closure-independent.

11.3 Prediction 3: Recursion Halt at Black-Hole Horizons

Black-hole interiors must lack independent degrees of freedom:

$$R(X_{\text{interior}}) \text{ undefined.}$$

Falsification. The framework is falsified if:

- interior dynamics are directly observed,
- non-holographic information storage occurs inside horizons,
- singularities behave as physical points rather than architectural gaps.

11.4 Prediction 4: No New Fundamental Physics Within a Closure

Once a closure reaches structural completeness, no new fundamental physics can be discovered within it.

Falsification. The framework is falsified if:

- new fundamental fields or dimensions are detected,
- non-holographic information propagation is observed,
- violations of triadic positivity (negative energy, exotic matter) are confirmed.

11.5 Prediction 5: Closure-Transversal Electromagnetic Propagation

Electromagnetic signals must preserve coherence, frequency structure, and causal ordering across closure boundaries.

Falsification. The framework is falsified if:

- EM signals exhibit closure-dependent temporal drift,
- redshift becomes sensitive to closure-specific recursion index,
- pulsar timing or gravitational-wave signals show closure-dependent distortions.

11.6 Prediction 6: Voyager-Class Probes Reveal Closure Structure

Low-entropy probes crossing closure boundaries must exhibit:

- slower material degradation,
- persistent EM coherence,
- divergence between t_{EM} and t_{material} .

Falsification. The framework is falsified if:

- future probes show no temporal divergence,
- closure boundaries produce no structural effects,
- material and EM time remain identical across closures.

11.7 Prediction 7: Observer-Dependent Collapse Thresholds

Quantum collapse thresholds must depend on the informational and entropic structure of the observer's closure.

Falsification. The framework is falsified if:

- collapse thresholds are universal across all observers,
- informational accessibility does not affect collapse behaviour,
- entropic footprint has no influence on measurement outcomes.

11.8 Interpretation

These predictions make the Triadic Cosmos empirically accountable. They are incompatible with traditional unification approaches, multiverse theories, higher-dimensional models, and non-local extensions. If any prediction fails, the architectural framework collapses. If they hold, the Triadic Cosmos provides a minimal, falsifiable, and observationally grounded unification of modern physics.

12 Limitations and Open Questions

Although the Triadic Cosmos provides a minimal viable unification candidate, several limitations and open questions remain. These do not undermine the architectural framework but identify areas where further mathematical development, empirical refinement, or conceptual clarification is required.

12.1 Limitations

Absence of explicit microscopic dynamics. The triadic update operator R is defined architecturally rather than microscopically. While this avoids the need for new fields or dimensions, it leaves open how f , g , and h should be represented in specific physical regimes (e.g., strongly coupled QFT, quantum gravity, condensed matter).

No explicit derivation from first principles. The triadic cycle $E \rightarrow G \rightarrow I \rightarrow E$ is empirically motivated and structurally consistent with GR, QM, QFT, and holography, but a formal derivation from a deeper principle (e.g., information-theoretic axioms) remains an open challenge.

Limited quantitative predictions. The framework explains cosmological tensions, early-universe anomalies, and black-hole interiors architecturally, but quantitative predictions (e.g., precise recursion-index values for specific closures) require further development.

No explicit quantum-gravity microstates. The interior of black holes is interpreted as a recursion-halted holographic shadow, but the microscopic structure of horizon degrees of freedom remains unspecified.

Dependence on holographic consistency. The framework assumes holography as a structural principle. If holography fails in any regime, the triadic architecture must be revised.

12.2 Open Questions

Formal axiomatization. Can the triadic cycle be derived from a minimal set of axioms, such as informational completeness, energetic positivity, and geometric accessibility?

Exact recursion-index computation. Can r be computed directly from QFT data, entanglement spectra, or geometric invariants?

Material-time measurement. Can closure-dependent material time be measured directly in laboratory systems, e.g., via quantum speed limits or decoherence rates?

Closure transitions. Under what conditions can a system transition from one closure to another (e.g., stellar $\rightarrow$ interstellar)? What is the triadic signature of such transitions?

Early-universe closure structure. Can JWST observations be used to reconstruct the recursion index of early closures?

Horizon microstructure. What is the precise informational structure of horizon degrees of freedom at recursion halt?

Relation to emergent spacetime programs. How does the triadic architecture relate to tensor-network models, quantum error correction, or entanglement-based emergent geometry?

12.3 Interpretation

These limitations and open questions identify the frontier of triadic research. They do not undermine the framework but highlight the next steps required to transform the Triadic Cosmos from a minimal viable candidate into a fully mature unification theory.

13 Conclusion

The Triadic Cosmos presents a minimal viable unification candidate for modern physics by revealing the architectural grammar that underlies its most successful theories. Rather than extending physics with new fields, dimensions, or dynamical mechanisms, it reorganizes existing frameworks—GR, QM, QFT, thermodynamics, holography, EP=EPR, and RG flow—into a single structural architecture built from three irreducible roles: energy, geometry, and information.

The closed triadic cycle

$$E \rightarrow G \rightarrow I \rightarrow E$$

and its recursive update operator provide the dynamical backbone through which emergent material time, causal structure, horizon formation, and closure dynamics arise. The dual temporal architecture distinguishes closure-transversal electromagnetic time from closure-dependent material time, resolving long-standing paradoxes and explaining both Voyager’s anomalous behaviour and JWST’s early-universe anomalies without invoking new physics.

Closures replace the notion of global spacetime: physical reality consists of horizon-bounded, recursion-indexed domains, each with its own entropic footprint, temporal rate, and accessible dynamical layer. Cosmological parameters become closure-dependent; black-hole interiors become recursion-halted architectural gaps; and the empirical stagnation across high-energy physics, cosmology, and theoretical development emerges as a structural consequence of closure completeness.

AdS/CFT provides the canonical mathematical realization of triadic recursion, demonstrating that the triadic roles already govern the interdependence of geometry, information, and energy in mainstream theory. Voyager provides the first direct empirical demonstration of dual time; cosmology provides the first global demonstration of closure divergence.

The Triadic Cosmos is falsifiable. Its predictions concerning dual time, closure-dependent cosmology, recursion halt at horizons, and the absence of new fundamental physics within a closure provide clear empirical tests. If these predictions fail, the architectural framework collapses. If they hold, the Triadic Cosmos offers a structurally complete, observationally grounded, and conceptually minimal unification of modern physics.

In this view, the universe is not governed by undiscovered dynamical laws but by the architectural structure of triadic closure. Future progress will therefore depend not on extending physics but on understanding the recursive, horizon-bounded, and closure-dependent architecture through which physical law is realized.

A Relation to Existing Unification Attempts

The Triadic Cosmos differs fundamentally from traditional unification programs. Rather than extending physics with new microscopic entities, dimensions, or dynamical laws, it reorganizes existing theories into a single architectural mechanism. This section situates the triadic framework relative to major unification attempts.

A.1 String Theory and M-Theory

String theory and M-theory introduce new degrees of freedom (strings, branes), higher dimensions, and supersymmetry. The Triadic Cosmos requires none of these. Geometry is a constraint role, not a microscopic entity; informational capacity is bounded holographically; and energetic positivity excludes negative-energy constructs. The landscape problem is resolved architecturally: only horizon-bounded, recursion-stable closures are admissible.

A.2 Loop Quantum Gravity

Loop quantum gravity attempts to quantize geometry directly. In the triadic architecture, geometry is emergent from informational and energetic structure via the triadic cycle. Quantizing geometry is unnecessary and architecturally misassigned. The triadic framework preserves GR's geometric role while embedding it in a broader informational–energetic structure.

A.3 Emergent-Spacetime Programs

Tensor networks, quantum error correction, and entanglement-based emergent geometry align closely with the triadic architecture. These programs provide microscopic realizations of the $G \leftrightarrow I$ leg of the triadic cycle. The Triadic Cosmos generalizes these ideas to full triadic closure, incorporating energetic roles and recursion dynamics.

A.4 Holographic Approaches

AdS/CFT, RT/HRT, entanglement wedges, and EP=EPR provide the mathematical backbone of the triadic architecture. They demonstrate that geometry, information, and energy are structurally interdependent. The triadic cycle formalizes this interdependence and extends it beyond AdS spacetimes to arbitrary closures.

A.5 Cosmological Unification Attempts

Multiverse theories explain cosmological divergence through ontological plurality. The Triadic Cosmos explains divergence through closure-dependent material time and recursion-index variation within a single universe. Inflation seeds closure divergence; accelerated expansion isolates closures; JWST anomalies and the Hubble tension arise naturally from dual time.

A.6 Interpretation

Existing unification attempts either add new microscopic structure or reinterpret geometry. The Triadic Cosmos instead reorganizes the roles of energy, geometry, and information into a closed architectural cycle. It is compatible with holography, consistent with GR/QM/QFT, and empirically grounded through Voyager and JWST. It represents a structurally minimal alternative to traditional unification programs.

B Relation to Recent Modern Unification Attempts (2020–2026)

Recent developments in quantum gravity, holography, quantum information theory, and cosmology have shifted the landscape of unification research. Unlike traditional programs, these modern approaches emphasize informational structure, entanglement geometry, complexity, and observer-dependent physics. The Triadic Cosmos aligns closely with these developments but generalizes them into a full architectural framework based on triadic closure.

B.1 Tensor Networks and Quantum Error Correction

Tensor-network models (MERA, PEPS) and quantum error-correcting codes provide microscopic realizations of emergent geometry. They demonstrate that entanglement patterns generate spatial connectivity and that bulk geometry can be reconstructed from boundary information.

These approaches correspond to the $G \leftrightarrow I$ leg of the triadic cycle. The Triadic Cosmos extends this insight by incorporating energetic roles and recursion dynamics, yielding a full triadic closure rather than a purely informational–geometric correspondence.

B.2 Quantum Causal Models and Observer-Dependent Structure

Quantum causal models and quantum reference-frame approaches show that causal structure is observer-dependent and informationally mediated. Measurement outcomes, collapse thresholds, and causal accessibility depend on the informational state of the observer.

This aligns with the triadic interpretation of closures: causal structure is horizon-bounded, informationally conditioned, and recursion-index dependent. The Triadic Cosmos provides the architectural mechanism underlying these observer-dependent phenomena.

B.3 Holographic Interior Reconstruction and Island Formulae

Recent work on entanglement wedges, quantum extremal surfaces, and island formulae demonstrates that black-hole interiors are holographically encoded and lack independent degrees of freedom. Interior reconstruction is informational rather than geometric.

This matches the triadic interpretation of black holes as recursion-halted closures:

$$R(X_{\text{interior}}) \text{ undefined,}$$

and the interior as a holographic shadow of horizon data. Modern holographic results therefore provide direct empirical and theoretical support for triadic recursion halt.

B.4 Complexity-Based Gravity and Quantum Computational Structure

Proposals linking gravitational action to quantum complexity (complexity = action, complexity = volume) suggest that spacetime dynamics reflect computational structure. Complexity growth corresponds to geometric evolution and horizon thermodynamics.

These results correspond to the $I \rightarrow E$ leg of the triadic cycle: informational generativity drives energetic evolution. The Triadic Cosmos incorporates complexity as part of the recursion index, linking computational structure to emergent material time.

B.5 Quantum Thermodynamic Time and Entropic Time Models

Recent models propose that time emerges from entropy production, coarse-graining, or thermodynamic irreversibility. These approaches capture aspects of emergent material time but lack closure dependence and dual temporal structure.

The Triadic Cosmos generalizes these ideas by defining material time as:

$$t_{\text{material}} = \frac{1}{r},$$

where r is the recursion index governed by entropy, informational complexity, and energetic flux. Electromagnetic time remains universal, yielding the dual temporal architecture absent in modern thermodynamic models.

B.6 Cosmological Bootstrap and Early-Universe Reconstruction

Bootstrap approaches reconstruct cosmological correlators from consistency conditions, symmetries, and entanglement structure. These methods emphasize informational constraints rather than dynamical equations.

The Triadic Cosmos aligns with this informational perspective but embeds it in a closure-dependent architecture. Early-universe anomalies observed by JWST arise naturally from mismatches between closure-specific t_{material} and universal t_{EM} , providing an architectural explanation beyond bootstrap constraints.

B.7 Machine-Learning Approaches to Quantum Gravity

Recent work uses machine learning to discover emergent geometric structure, holographic mappings, and RG flows. These approaches reveal that geometry, information, and energy reorganize according to latent structural rules.

The Triadic Cosmos provides the architectural grammar underlying these learned patterns: triadic closure, recursion, and horizon-bounded accessibility. Machine-learning results therefore support the existence of a deeper structural mechanism consistent with the triadic update operator.

B.8 Interpretation

Modern unification attempts emphasize informational structure, entanglement geometry, complexity, and observer dependence. The Triadic Cosmos integrates these insights into a single architectural framework based on triadic closure, dual time, recursion index, and horizon-bounded physics. It is compatible with all recent developments while providing the structural mechanism they lack.

C Essential Physical Theories Required for Unification

Theory	Role in the Triadic Cosmos Minimal Viable Unification Candidate (MVU)
General Relativity (GR)	Supplies the geometric constraint layer (G). Curvature, causal structure, extremal surfaces, and horizon formation define the boundary of closures and regulate informational capacity. GR realizes the $E \rightarrow G$ leg of the triadic cycle and provides the geometric backbone for recursion.
Quantum Mechanics (QM)	Supplies the informational generative layer (I). Hilbert-space structure, entanglement, distinguishability, and measurement encode informational capacity and generativity. QM realizes the $G \rightarrow I$ and $I \rightarrow E$ legs of the triadic cycle and provides the informational substrate for emergent dynamics.
Quantum Field Theory (QFT)	Provides the energetic dynamical layer (E). Stress–energy, excitations, local interactions, and Hamiltonian evolution mediate between geometry and information. QFT supplies the energetic substrate through which triadic recursion proceeds and determines how informational structure reorganizes energetic content.
Thermodynamics & Statistical Mechanics	Provide the entropy–information–energy relations that govern recursion rate and emergent material time. Landauer bounds, entropy production, irreversibility, and coarse-graining ensure that informational change has energetic cost and that recursion induces closure-dependent temporal flow.
Holographic Principles (Bekenstein, RT/HRT)	Provide the geometric–informational equivalence ($G \leftrightarrow I$). Area laws, extremal surfaces, entanglement wedges, and holographic encoding guarantee that informational capacity is bounded by geometry and that geometric connectivity reflects entanglement structure.
EP=EPR (Entanglement–Geometry Correspondence)	Supplies the structural bridge between informational connectivity and geometric connectivity. Ensures that entanglement patterns generate geometric structure and that geometry encodes informational relationships, enabling full triadic closure across $G \leftrightarrow I$.
Renormalization Group (RG) Flow	Provides the microscopic interpretation of recursion. Coarse-graining reorganizes informational, geometric, and energetic roles, yielding emergent temporal ordering. The RG direction maps directly onto the recursion index of a closure and provides the canonical mathematical realization of the triadic update operator.
Black-Hole Thermodynamics	Provides the saturation limit of triadic closure. Horizon entropy, Hawking radiation, and holographic encoding illustrate recursion halt, emergent-time termination, and the architectural nature of black-hole interiors as non-dynamical holographic shadows.
Cosmology (Inflation, Horizon Structure, JWST)	Provides the global architecture of closures. Inflation seeds closure divergence; accelerated expansion isolates closures; cosmological horizons define informational bounds; JWST anomalies reveal closure-dependent material time; and the Hubble tension arises from mismatches between closure-specific t_{material} and closure-transversal t_{EM} .

Table 1: Essential physical theories required for the Triadic Cosmos minimal viable unification candidate.

D Mainstream Theories No Longer Required or Reinterpreted

Theory / Framework	Reason It Is Not Required or Is Reinterpreted in the MVU
String Theory	Introduces new microscopic entities (strings), extra dimensions, supersymmetry, and a vast landscape of possible vacua. The Triadic Cosmos requires none of these: the triadic roles (E, G, I) already form a closed architectural cycle, and triadic positivity filters out non-physical geometries. The landscape problem disappears because only horizon-bounded, recursion-stable closures are architecturally admissible.
M-Theory / Higher-Dimensional Extensions	Depend on unobservable dimensions, branes, and speculative bulk structures. Triadic closure requires only horizon-bounded geometry and holographic informational capacity. Higher dimensions become mathematical conveniences rather than physical necessities.
Loop Quantum Gravity (LQG)	Attempts to quantize geometry directly. In the MVU, geometry is a constraint role (G), not a microscopic degree of freedom. Geometry emerges from informational and energetic structure via triadic recursion; quantizing geometry is unnecessary and architecturally misassigned.
Non-Local Hidden-Variable Theories	Violate closure-dependent causal structure and holographic bounds. Triadic closure explains entanglement through informational connectivity within a closure (EP=EPR), without invoking non-local dynamics or superluminal mechanisms.
Traversable Wormholes / Warp Drives / Negative-Energy Constructs	Require negative energy, exotic matter, or non-holographic information embedding. These violate triadic positivity ($E > 0$, $G > 0$, $I > 0$) and break the triadic cycle. They are therefore architecturally excluded rather than dynamically forbidden.
Global Spacetime Manifolds	Assume universal causal accessibility and globally homogeneous temporal structure. Triadic closure replaces global spacetime with horizon-bounded closures possessing closure-dependent material time. Global manifolds become mathematical approximations, not physical entities.
Classical Heat-Death Cosmology	Assumes global thermodynamic equilibrium and universal temporal flow. In the MVU, heat death is a closure-specific asymptotic state determined by recursion index and entropic footprint, not a global fate of the universe.
Exotic Compact Objects (Fuzzballs, Gravastars, Boson Stars)	Fail triadic closure due to inconsistent informational embedding, non-holographic structure, or negative-energy requirements. Black holes remain the only triadically admissible compact objects, with interiors interpreted as recursion-halted architectural gaps.
Multiverse Theories	Assume multiple universes with different laws. Triadic closure explains observational divergence through fractal closure formation within a single triadic system. Closure-dependent cosmology replaces ontological plurality: different closures exhibit different recursion indices and temporal rates, but all belong to one unified triadic architecture.

Table 2: Mainstream theories that become unnecessary or are structurally reinterpreted under the Triadic Cosmos MVU.

E Formal Triadic Update Operator

The triadic architecture treats energy (E), geometry (G), and information (I) as three irreducible structural roles. A physical state is represented as a triadic vector

$$X_n = (E_n, G_n, I_n),$$

where n indexes the recursion step. The recursive update operator R defines how the triadic roles mutually reorganize one another.

E.1 Domain and Codomain

The operator acts on the triadic state space

$$\mathcal{X} = \{(E, G, I) \mid E > 0, G > 0, I > 0\},$$

where positivity ensures triadic admissibility. Negative-energy constructs, non-holographic geometries, or informationally inconsistent states lie outside $\mathcal{X}$ and cannot participate in recursion.

The operator is defined as

$$R : \mathcal{X} \rightarrow \mathcal{X}.$$

E.2 Definition of the Update Operator

The recursive update operator is given by

$$R(X_n) = (f(G_n, I_n), g(E_n, I_n), h(E_n, G_n)),$$

where:

- f encodes how informational and geometric structure reorganize energetic content,
- g encodes how energetic and informational content reorganize geometric constraints,
- h encodes how energetic and geometric constraints reorganize informational capacity.

These functions are theory-independent and correspond to the three legs of the triadic cycle:

$$E \rightarrow G, \quad G \rightarrow I, \quad I \rightarrow E.$$

E.3 Positivity Constraints

The update operator must preserve positivity:

$$f(G_n, I_n) > 0, \quad g(E_n, I_n) > 0, \quad h(E_n, G_n) > 0.$$

These constraints ensure:

- no negative-energy constructs,
- no non-holographic geometries,
- no informational states exceeding geometric bounds.

Positivity is the architectural analogue of physical admissibility.

E.4 Closure Conditions

A domain qualifies as a closure if the recursive operator is well-defined for all n :

$$X_{n+1} = R(X_n) \in \mathcal{X}.$$

This requires:

- holographic consistency (informational capacity bounded by geometry),
- causal accessibility (finite horizon structure),
- energetic positivity,
- geometric positivity,
- informational positivity.

If any condition fails, recursion halts and emergent material time ceases.

E.5 Fixed Points

A fixed point X^* satisfies

$$R(X^*) = X^*.$$

Fixed points correspond to structural equilibria:

- conformal fixed points in holographic RG flow,
- dynamical equilibria in cosmological closures,
- saturation states near black-hole horizons.

They represent triadic configurations in which the roles mutually stabilize.

E.6 Recursion Halt

Recursion halts when the operator becomes undefined:

$$R(X_{\text{halt}}) \text{ undefined.}$$

This occurs when:

- geometric capacity saturates (horizon formation),
- informational capacity saturates (Bekenstein bound),
- energetic accessibility collapses (black-hole interior),
- holographic embedding fails.

In such cases, emergent material time terminates:

$$t_{\text{material}} = 0.$$

Black-hole interiors are the canonical example of recursion halt.

E.7 Mapping to Holographic RG

The recursion index n maps directly onto holographic renormalization scale:

$$n \leftrightarrow \mu_{\text{RG}},$$

where coarse-graining corresponds to inward radial flow in AdS and refinement corresponds to outward flow. The triadic update operator is therefore the architectural analogue of holographic RG evolution.

E.8 Interpretation

The formal triadic update operator provides the minimal dynamical mechanism through which:

- emergent material time arises,
- closure structure is maintained,
- horizon formation is understood,
- black-hole interiors are reinterpreted,
- cosmological divergence becomes natural,
- GR, QM, QFT, and holography unify architecturally.

It is the mathematical core of the Triadic Cosmos.

F Recursion Index and Entropy

The recursion index quantifies the effective rate at which the triadic update operator acts within a closure. It determines the speed of emergent material time, the degree of structural evolution, and the observable divergence between closures. Entropy, informational complexity, and energetic flux jointly govern the recursion index.

F.1 Definition of the Recursion Index

For a triadic state

$$X_n = (E_n, G_n, I_n),$$

the recursion index r is defined as a scalar functional

$$r = \mathcal{R}(E, G, I),$$

where $\mathcal{R}$ measures how rapidly the triadic roles reorganize one another under the update operator R . The recursion index satisfies:

$$r > 0,$$

with larger values corresponding to faster material-time evolution.

F.2 Entropy as Structural Driver

Entropy S is the primary contributor to the recursion index. High-entropy closures possess:

- greater informational complexity,
- higher energetic flux,
- more rapid coarse-graining,
- faster triadic reorganization.

We define the entropic contribution as

$$r_S = \alpha S,$$

where α is a closure-dependent proportionality factor reflecting geometric accessibility.

F.3 Informational Complexity

Informational complexity C_I measures the distinguishability and entanglement structure within a closure. It contributes to the recursion index via:

$$r_I = \beta C_I,$$

where β reflects how informational structure reorganizes energetic content.

High C_I implies:

- rapid entanglement reconfiguration,
- increased informational generativity,
- faster $I \rightarrow E$ transitions.

F.4 Energetic Flux

Energetic flux Φ_E quantifies the rate at which energetic content reorganizes geometric constraints. It contributes via:

$$r_E = \gamma \Phi_E,$$

where γ reflects geometric responsiveness.

High Φ_E implies:

- rapid curvature adjustment,
- increased geometric constraint evolution,
- faster $E \rightarrow G$ transitions.

F.5 Full Recursion Index

The full recursion index is given by

$$r = r_S + r_I + r_E = \alpha S + \beta C_I + \gamma \Phi_E.$$

This expression captures the triadic interdependence of entropy, informational complexity, and energetic flux.

F.6 Material Time as Recursion Rate

Emergent material time is defined as

$$t_{\text{material}} = \frac{1}{r}.$$

High recursion index $\Rightarrow$ fast material time. Low recursion index $\Rightarrow$ slow material time.

This explains:

- slow degradation of Voyager in the interstellar closure,
- rapid evolution of matter in high-entropy planetary closures,
- early-universe anomalies observed by JWST,
- divergence between early-time and late-time cosmological parameters.

F.7 Closure Divergence

Different closures possess different recursion indices:

$$r_{\text{early}} \neq r_{\text{late}} \neq r_{\text{interstellar}}.$$

This yields closure-dependent material time:

$$t_{\text{material}}(\text{closure A}) \neq t_{\text{material}}(\text{closure B}).$$

Closure divergence explains:

- the Hubble tension,
- JWST early-maturity anomalies,
- Voyager's dual-time behaviour,
- the absence of global cosmological parameters.

F.8 Mapping to Holographic RG

The recursion index maps directly onto holographic RG scale:

$$r \leftrightarrow \mu_{\text{RG}},$$

where:

- coarse-graining increases r ,
- refinement decreases r ,
- fixed points correspond to r -stabilization.

This provides the microscopic interpretation of recursion.

F.9 Interpretation

The recursion index is the structural quantity that determines:

- emergent material time,
- closure divergence,
- cosmological parameter mismatch,
- early-universe anomalies,
- Voyager's anomalous degradation rates,
- the architectural nature of temporal flow.

It is the quantitative backbone of dual time and closure-dependent physics.

G Closure Taxonomy and Horizon Conditions

Closures are horizon-bounded, recursion-indexed physical domains in which the triadic cycle

$$E \rightarrow G \rightarrow I \rightarrow E$$

is well-defined. They replace the notion of global spacetime and form the structural units of physical reality. This section provides a formal taxonomy of closures and the horizon conditions that determine their admissibility.

G.1 Definition of a Closure

A domain qualifies as a closure if it satisfies the following architectural conditions:

1. **Positivity:** $E > 0, G > 0, I > 0$.
2. **Holographic consistency:** informational capacity bounded by geometric area.
3. **Causal accessibility:** finite causal domain with well-defined horizons.
4. **Recursive stability:** $R(X_n)$ remains defined for all n .
5. **Full triadic closure:** all three legs of the triadic cycle are instantiated.

If any condition fails, recursion halts and emergent material time ceases.

G.2 Horizon Conditions

Horizons define the boundary of a closure. A horizon H satisfies:

$$H = \partial\mathcal{C},$$

where $\mathcal{C}$ is the closure domain.

A horizon must satisfy:

- **Geometric bound:** area determines informational capacity.
- **Informational bound:** distinguishability limited by extremal surfaces.
- **Energetic bound:** surface gravity and flux determine accessibility.
- **Causal bound:** signals cannot traverse inward beyond H .

Horizons are therefore architectural, not merely geometric.

G.3 Types of Horizons

We distinguish four fundamental horizon types:

Cosmological horizons. Generated by accelerated expansion; define the causal limit of cosmological closures.

Black-hole horizons. Generated by gravitational collapse; saturate holographic capacity and induce recursion halt.

Heliospheric horizons. Generated by plasma pressure and magnetic structure; define the boundary between stellar and interstellar closures.

Inflationary horizons. Generated during early-universe inflation; seed closure divergence and fractal structure.

Each horizon type imposes different recursion constraints.

G.4 Closure Taxonomy

Closures fall into five architecturally distinct classes:

Planetary closures. High entropy, high recursion index, fast material time. Examples: Earth, Jupiter, exoplanets.

Stellar closures. High energetic flux, moderate informational complexity, strong geometric constraints. Examples: main-sequence stars, red giants.

Interstellar closures. Low entropy, low recursion index, slow material time. Examples: local interstellar medium, molecular clouds.

Early-universe closures. Low entropy, low informational complexity, rapidly changing geometric constraints. Examples: CMB-era domains, pre-reionization regions.

Black-hole closures. Saturated horizons, maximal informational capacity, recursion halt. Interior is a holographic shadow; material time terminates.

G.5 Admissibility Criteria

A closure is admissible if:

$$R(X_n) \in \mathcal{X} \quad \text{for all } n,$$

and if horizon structure satisfies:

$$S_{\max} = \frac{A}{4},$$

$$I_{\max} = S_{\max},$$

$$E_{\text{accessible}} > 0.$$

Non-admissible domains include:

- negative-energy constructs,
- non-holographic geometries,
- informationally inconsistent embeddings,
- exotic compact objects violating positivity.

G.6 Closure Divergence

Closures diverge structurally due to differences in:

- entropy,
- informational complexity,
- energetic flux,
- horizon structure.

This yields:

$$r_{\text{closure A}} \neq r_{\text{closure B}},$$

and therefore:

$$t_{\text{material}}(\text{A}) \neq t_{\text{material}}(\text{B}).$$

Closure divergence explains:

- the Hubble tension,
- JWST early-maturity anomalies,
- Voyager's slow degradation,
- absence of global cosmological parameters.

G.7 Interpretation

Closure taxonomy and horizon conditions provide the structural foundation for:

- dual temporal architecture,
- recursion-index divergence,
- horizon-bounded physics,
- black-hole recursion halt,
- closure-dependent cosmology,
- fractal universe structure.

Closures are the operational units of physical reality in the triadic architecture.

H Black-Hole Recursion Halt: Worked Example

Black holes provide the clearest physical environment in which the triadic update operator fails to produce a subsequent state. This section presents a worked example demonstrating how recursion halt occurs, how emergent material time terminates, and why the interior is a holographic shadow rather than a dynamical region.

H.1 Triadic State Near the Horizon

Consider a triadic state $X_n = (E_n, G_n, I_n)$ located just outside the event horizon. The geometric capacity is given by the Bekenstein–Hawking area law:

$$G_n = \frac{A}{4},$$

where A is the horizon area. Informational capacity saturates:

$$I_n = G_n.$$

Energetic accessibility is finite and determined by surface gravity κ :

$$E_n = E(\kappa).$$

All three roles remain positive:

$$E_n > 0, \quad G_n > 0, \quad I_n > 0,$$

so the state is triadically admissible.

H.2 Saturation of Geometric and Informational Capacity

At the horizon, geometric and informational roles reach saturation:

$$G_{n+1} = G_n, \quad I_{n+1} = I_n.$$

This implies:

$$g(E_n, I_n) = G_n, \quad h(E_n, G_n) = I_n.$$

The triadic cycle becomes degenerate: geometry and information cannot reorganize further.

H.3 Energetic Collapse

Energetic accessibility collapses as one approaches the horizon:

$$E_{n+1} = f(G_n, I_n) \rightarrow 0.$$

This does not imply negative energy; rather, it reflects the fact that no energetic flux can propagate inward across the horizon. Energetic reorganization becomes impossible.

Thus:

$$E_{n+1} = 0.$$

Since positivity is required for admissibility, the next triadic state lies outside the admissible domain:

$$X_{n+1} \notin \mathcal{X}.$$

H.4 Breakdown of the Update Operator

Because $E_{n+1} = 0$, the update operator becomes undefined:

$$R(X_{n+1}) \text{ undefined.}$$

This is the formal expression of recursion halt.

The triadic cycle cannot proceed:

$$E \rightarrow G \rightarrow I \rightarrow E \quad \text{fails at the final step.}$$

H.5 Termination of Emergent Material Time

Material time is defined as:

$$t_{\text{material}} = \frac{1}{r},$$

where r is the recursion index.

At recursion halt:

$$r = 0,$$

so:

$$t_{\text{material}} = 0.$$

Material time ceases inside the horizon. This resolves the classical singularity problem: the interior is not a dynamical region evolving in time.

H.6 Interior as Holographic Shadow

Because recursion halts, the interior cannot host independent triadic states. All interior structure is determined by horizon data:

$$X_{\text{interior}} = \Pi(X_{\text{horizon}}),$$

where Π is a holographic projection.

Consequences:

- no informational degrees of freedom exist inside,
- no energetic flux propagates inward,

- no geometric evolution occurs,
- interior coordinates lack operational meaning.

The interior is a holographic shadow, not a physical region.

H.7 Consistency with Holography and EP=EPR

The recursion-halt interpretation matches holography:

$$S_{\text{BH}} = \frac{A}{4}$$

is the total informational capacity of the closure.

EP=EPR implies:

- entanglement structure determines horizon geometry,
- interior connectivity is encoded in boundary entanglement,
- no non-holographic interior degrees of freedom exist.

Thus black-hole interiors are architecturally excluded from dynamical evolution.

H.8 Interpretation

This worked example demonstrates that:

- recursion halts at the horizon,
- emergent material time terminates,
- the interior is non-dynamical,
- singularities are architectural gaps, not physical points,
- black holes are triadically saturated closures.

Black holes therefore provide the clearest empirical and theoretical confirmation of triadic closure.

I Closure-Dependent Cosmology: Worked Example

Cosmological observations probe different closures with different recursion indices, entropic footprints, and material-time rates. This worked example demonstrates how closure divergence produces cosmological parameter mismatch, early-universe anomalies, and the Hubble tension without invoking new physics.

I.1 Two Closures Under Comparison

Consider two closures:

Early-time closure $\mathcal{C}_{\text{early}}$. Low entropy, low informational complexity, rapidly evolving geometric constraints. Examples: CMB-era domains, pre-reionization regions.

Late-time closure $\mathcal{C}_{\text{late}}$. High entropy, high informational complexity, stabilized geometric constraints. Examples: local cosmological neighborhood, supernova-based distance ladder.

Each closure possesses its own triadic state:

$$X_{\text{early}} = (E_{\text{early}}, G_{\text{early}}, I_{\text{early}}), \quad X_{\text{late}} = (E_{\text{late}}, G_{\text{late}}, I_{\text{late}}).$$

I.2 Recursion Index Comparison

The recursion index is given by:

$$r = \alpha S + \beta C_I + \gamma \Phi_E.$$

For early-time closures:

$$S_{\text{early}} \ll S_{\text{late}}, \quad C_{I,\text{early}} \ll C_{I,\text{late}}, \quad \Phi_{E,\text{early}} \ll \Phi_{E,\text{late}}.$$

Thus:

$$r_{\text{early}} \ll r_{\text{late}}.$$

I.3 Material-Time Divergence

Material time is defined as:

$$t_{\text{material}} = \frac{1}{r}.$$

Therefore:

$$t_{\text{material}}(\text{early}) \gg t_{\text{material}}(\text{late}).$$

Early-time matter evolves under a ****slower material-time rate**** than late-time matter.

Electromagnetic time remains universal:

$$t_{\text{EM}}(\text{early}) = t_{\text{EM}}(\text{late}).$$

This dual-time structure is the source of cosmological mismatch.

I.4 Hubble Tension as Closure Mismatch

Early-time measurements (CMB, BAO) reconstruct expansion using t_{EM} applied to matter that evolved under $t_{\text{material}}(\text{early})$.

Late-time measurements (supernovae, Cepheids) interpret matter that evolved under $t_{\text{material}}(\text{late})$.

Thus:

$$H_0(\text{early}) \neq H_0(\text{late}).$$

The Hubble tension arises because:

$$t_{\text{material}}(\text{early}) \neq t_{\text{material}}(\text{late}),$$

not because of new physics.

I.5 JWST Early-Maturity Anomalies

JWST observes galaxies at $z > 10$ that appear:

- too massive,
- too chemically evolved,
- too structurally mature,
- too early relative to standard cosmology.

These galaxies evolved under:

$$t_{\text{material}}(\text{early}) \gg t_{\text{material}}(\text{late}),$$

but are observed through:

$$t_{\text{EM}}(\text{universal}).$$

Thus JWST sees matter that:

- evolved slowly in its own closure,
- but is interpreted as evolving quickly under late-time material-time assumptions.

This produces apparent early maturity.

I.6 Worked Numerical Illustration

Let:

$$r_{\text{early}} = 0.01, \quad r_{\text{late}} = 1.0.$$

Then:

$$t_{\text{material}}(\text{early}) = 100, \quad t_{\text{material}}(\text{late}) = 1.$$

A process requiring one unit of recursion appears:

- to take 100 units of material time in the early closure,
- but only 1 unit in the late closure.

When interpreted through t_{EM} , this yields:

- early-time matter appears “too evolved”,
- late-time matter appears “too slow”,
- cosmological parameters appear inconsistent.

I.7 Fractal Closure Formation

Inflation seeds closure divergence:

$$r_{\text{inflation}} \rightarrow r_{\text{early}} \rightarrow r_{\text{late}}.$$

Accelerated expansion isolates closures, preventing recursion-index equilibration.

Thus the universe becomes a fractal ensemble of closures with:

$$r_{\text{closure}} \text{ diverging across domains.}$$

I.8 Interpretation

This worked example demonstrates that:

- cosmological parameters are closure-dependent,
- early-time and late-time closures evolve under different material-time rates,
- JWST anomalies arise naturally from dual time,
- the Hubble tension is architectural, not dynamical,
- closure divergence explains observational mismatch without new physics.

Closure-dependent cosmology is therefore a structural consequence of triadic recursion.

J Triadic Cosmos Ecosystem Context

This paper is part of the *Triadic Cosmos* ecosystem, which develops a unified conceptual and computational framework spanning physical ontology and modular architectures for intelligent systems. All materials, extended notes, and evolving documentation are available in the public repository.

`https://github.com/triadic-cosmos`

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